problem1

May 22, 2026

0.1 3.1 Markovian drone

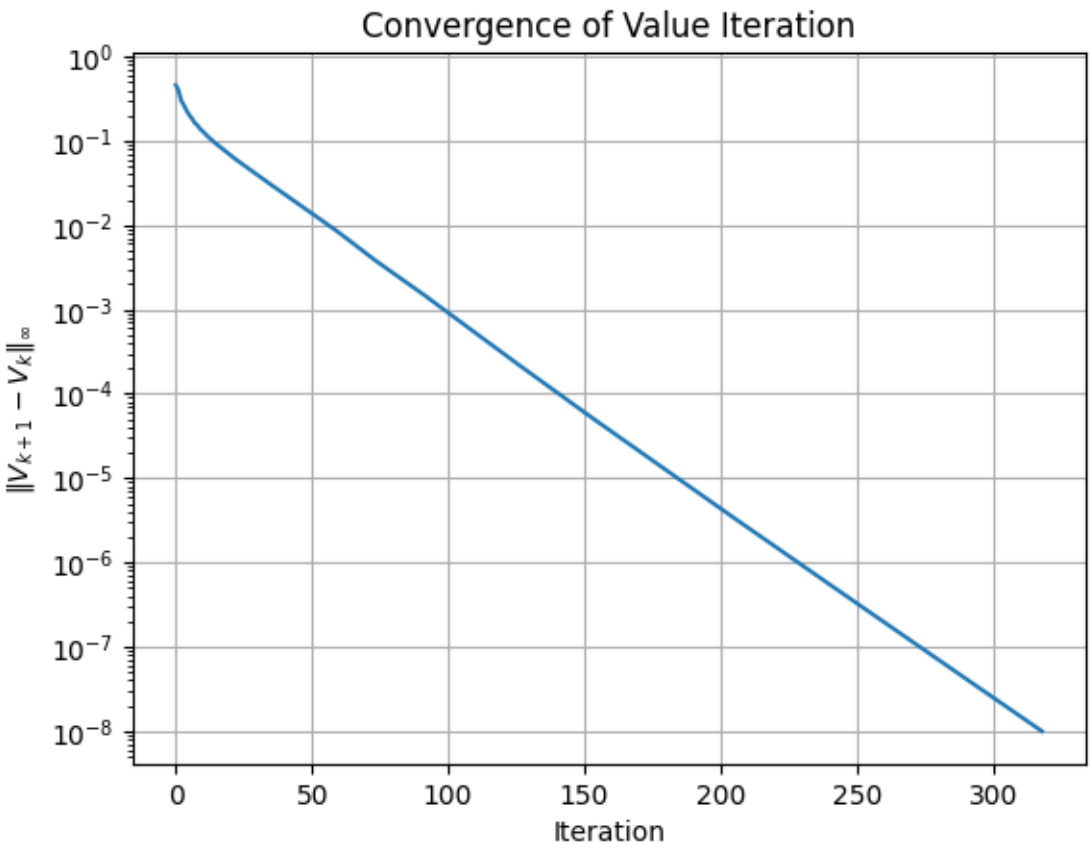
(a)

[1]:
```
import numpy as np
import matplotlib.pyplot as plt

n=20
drone_pos=np.array((0,0))
goal_pos=np.array((19,9))
x_eye=np.array((15,15))
sigma=10
gamma=0.95

actions={"up": np.array((0,1)),
         "down": np.array((0,-1)),
         "left": np.array((-1,0)),
         "right": np.array((1,0))}


def move_drone(pos,action):
    new_pos=pos+actions[action]
    if (0<=new_pos[0]<n) and (0<=new_pos[1]<n):
        return new_pos
    else:
        return pos

def get_reward(pos):
    return float(np.all(pos==goal_pos))

def get_neighboring_positions(pos):
    neighbors={}
    for action in actions.keys():
        neighbors[action]=move_drone(pos,action)
    return neighbors

def w(pos,x_eye):
    distance=np.linalg.norm(pos-x_eye)
```

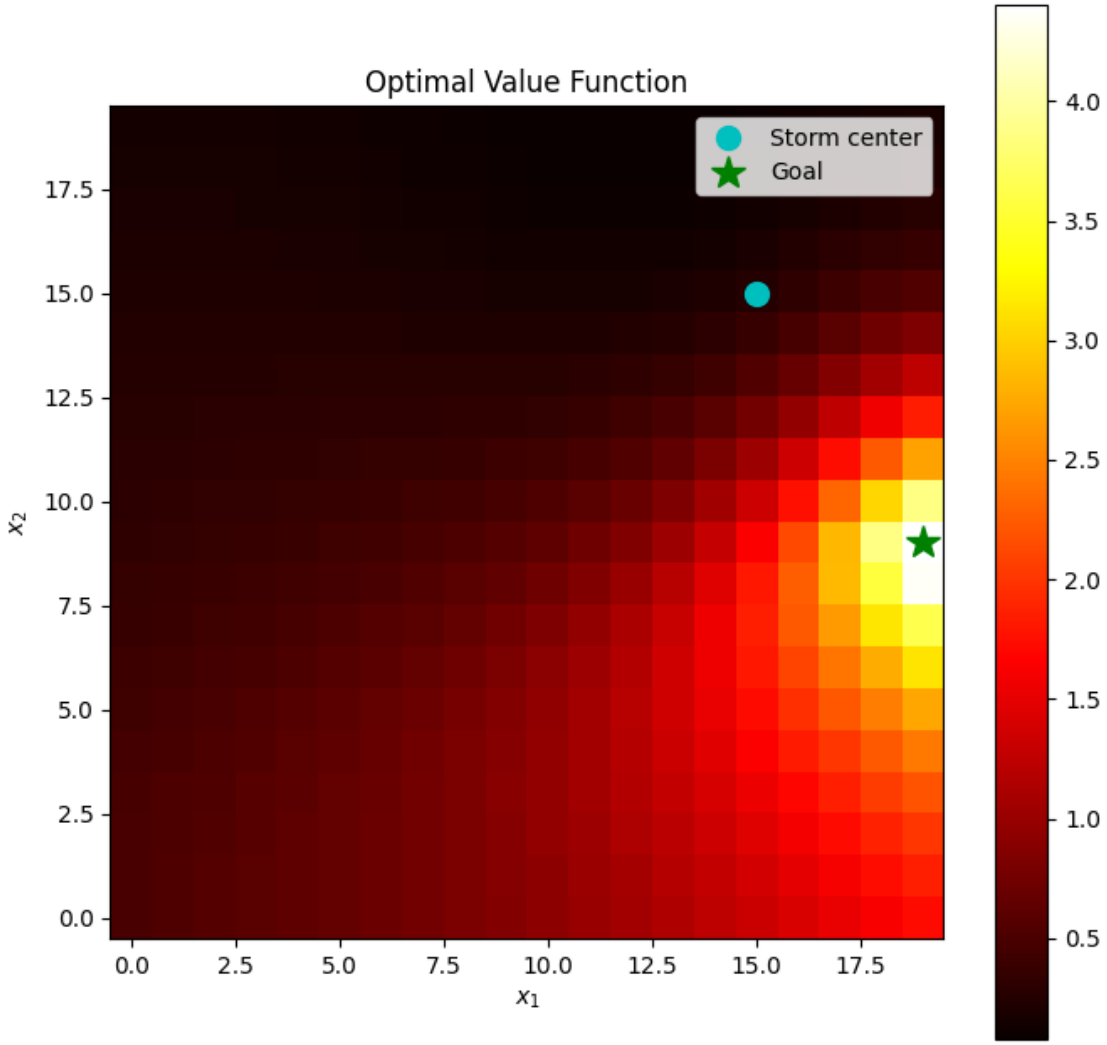
```
    return np.exp(-distance**2/(2*sigma**2))

def is_intended_action(action,intent):
    return action==intent

def bellman_update(V):
    V_new=np.zeros_like(V)
    for x in range(n):
        for y in range(n):
            pos=np.array((x,y))
            storm_prob=w(pos,x_eye)
            neighbors=get_neighboring_positions(pos)
            best_value=-np.inf
            for intent in actions.keys():
                value_candidate=sum(
                    (storm_prob/4 +␣
 ↪(1-storm_prob)*is_intended_action(a,intent))*(get_reward(neighbors[a]) +␣
 ↪gamma*V[tuple(neighbors[a])])
                    for a in actions.keys()
                )
                if value_candidate>best_value:
                    best_value=value_candidate
            V_new[x,y]=best_value
    return V_new
```

[2]:
```
V = np.zeros((n, n))
history = []
for i in range(500):
    V_new = bellman_update(V)
    diff = np.max(np.abs(V_new - V))
    history.append(diff)
    V = V_new
    if diff < 1e-8:
        break

print(f"Converged in {len(history)} iterations, peak V* = {V.max():.6f}")

plt.plot(history)
plt.yscale('log')
plt.xlabel('Iteration')
plt.ylabel(r'$\|V_{k+1} - V_k\|_\infty$')
plt.title('Convergence of Value Iteration')
plt.grid()
plt.show()
```

```
Converged in 319 iterations, peak V* = 4.402236
```



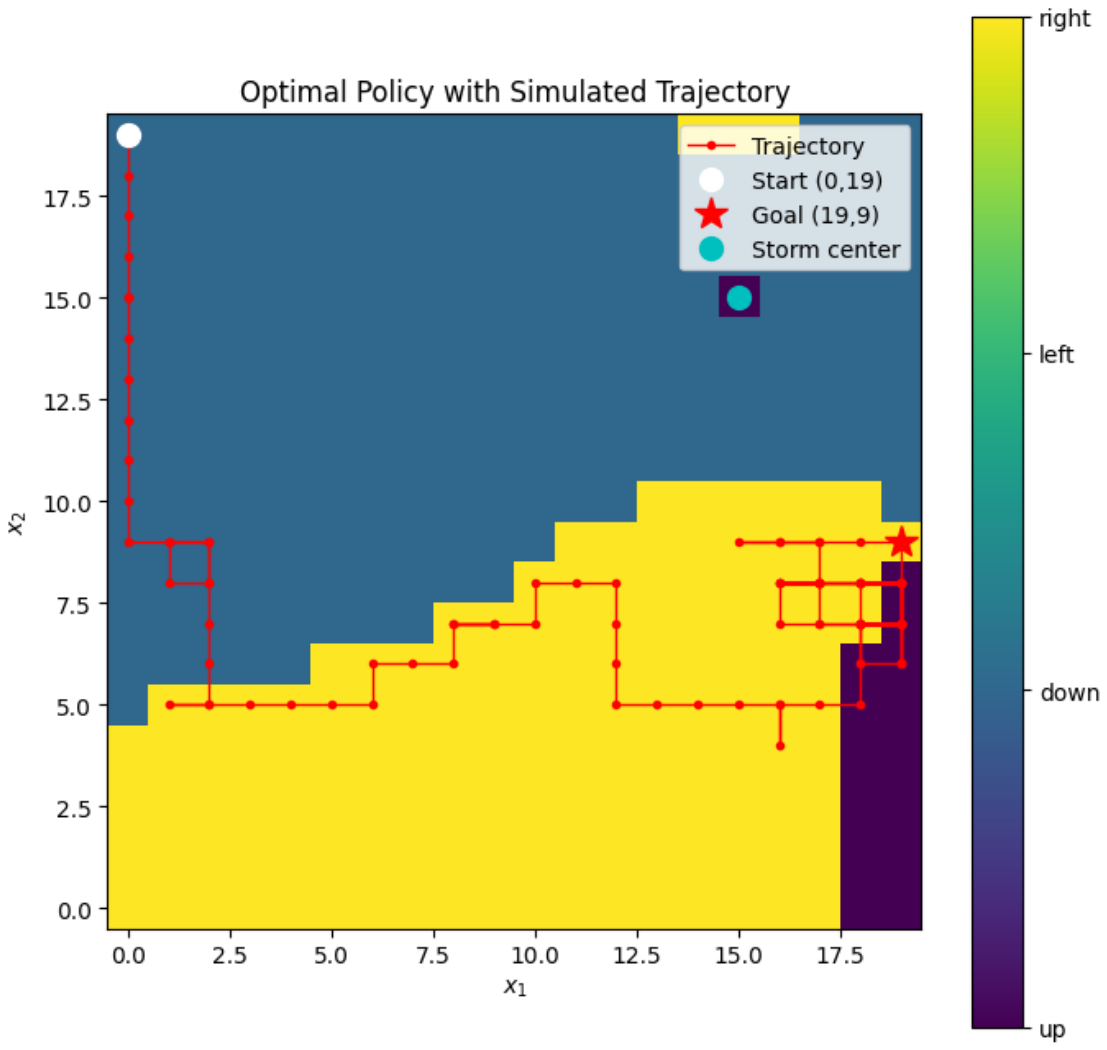
[3]:
```
fig, ax = plt.subplots(figsize=(8, 8))
im = ax.imshow(V.T, origin='lower', cmap='hot')
ax.set_xlabel('$x_1$')
ax.set_ylabel('$x_2$')
ax.set_title('Optimal Value Function')
ax.plot(x_eye[0], x_eye[1], 'co', markersize=10, label='Storm center')
ax.plot(goal_pos[0], goal_pos[1], 'g*', markersize=15, label='Goal')
ax.legend()
plt.colorbar(im, ax=ax)
plt.show()
```



(b)

[4]:
```
action_names = list(actions.keys())
policy = np.zeros((n, n), dtype=int)

for x in range(n):
    for y in range(n):
        pos = np.array((x, y))
        storm_prob = w(pos, x_eye)
        neighbors = get_neighboring_positions(pos)
        best_value = -np.inf
        for idx, intent in enumerate(action_names):
            value_candidate = sum(
                (storm_prob/4 + (1 - storm_prob) * is_intended_action(a,␣
 ↪intent)) *
```

```
                (get_reward(neighbors[a]) + gamma * V[tuple(neighbors[a])])
                for a in action_names
            )
            if value_candidate > best_value:
                best_value = value_candidate
                policy[x, y] = idx

# Simulate from (0, 19) for 100 steps
np.random.seed(42)
pos = np.array([0, 19])
trajectory = [pos.copy()]
for _ in range(100):
    action = action_names[policy[pos[0], pos[1]]]
    omega = w(pos, x_eye)
    if np.random.rand() < omega:
        action = np.random.choice(action_names)
    pos = move_drone(pos, action)
    trajectory.append(pos.copy())
trajectory = np.array(trajectory)

fig, ax = plt.subplots(figsize=(8, 8))
im = ax.imshow(policy.T, origin='lower', cmap='viridis')
ax.plot(trajectory[:, 0], trajectory[:, 1], 'r-o', markersize=3, linewidth=1,␣
 ↪label='Trajectory')
ax.plot(trajectory[0, 0], trajectory[0, 1], 'wo', markersize=10, label='Start␣
 ↪(0,19)')
ax.plot(goal_pos[0], goal_pos[1], 'r*', markersize=15, label='Goal (19,9)')
ax.plot(x_eye[0], x_eye[1], 'co', markersize=10, label='Storm center')
ax.set_xlabel('$x_1$')
ax.set_ylabel('$x_2$')
ax.set_title('Optimal Policy with Simulated Trajectory')
cbar = plt.colorbar(im, ax=ax, ticks=[0, 1, 2, 3])
cbar.set_ticklabels(['up', 'down', 'left', 'right'])
ax.legend()
plt.show()
```



The policy steers the drone toward the goal at $(19, 9)$ while routing around the storm center at $(15, 15)$. Far from the storm, the drone moves directly toward the goal. Near the eye, $\omega(x)$ is high, so any intended action has a large chance of being overridden by a uniformly random kick; in expectation that random kick costs more discounted reward than the longer detour, so the optimal policy routes around the high-ω region from below rather than through it.

Problem2

May 22, 2026

0.1 3.2 Reach-avoid flight

(a)

We want to find $\mathbf{u}^* = \arg\min_{\mathbf{u}} \mathbf{p}^\top f(\mathbf{x}, \mathbf{u})$ where $\mathbf{p} = \nabla_{\mathbf{x}} V$ and the reduced dynamics are

$$f(\mathbf{x}, \mathbf{u}) = \begin{bmatrix} v_y \\ \frac{(T_1+T_2)\cos\phi - C_D^v v_y}{m} - g \\ \omega \\ \frac{(T_2-T_1)\ell - C_D^\phi \omega}{I_{yy}} \end{bmatrix}.$$

Expanding $\mathbf{p}^\top f$, only the terms involving T_1, T_2 matter for the minimization:

$$\mathbf{p}^\top f = \cdots + T_1\left(\frac{p_2\cos\phi}{m} - \frac{p_4\ell}{I_{yy}}\right) + T_2\left(\frac{p_2\cos\phi}{m} + \frac{p_4\ell}{I_{yy}}\right).$$

This is linear in T_1 and T_2 (which are decoupled), so the minimum over $[0, T_{\max}]$ is attained at the extremes:

$$T_1^* = \begin{cases} T_{\max} & \text{if } \frac{p_2\cos\phi}{m} - \frac{p_4\ell}{I_{yy}} < 0, \\ 0 & \text{otherwise,} \end{cases} \qquad T_2^* = \begin{cases} T_{\max} & \text{if } \frac{p_2\cos\phi}{m} + \frac{p_4\ell}{I_{yy}} < 0, \\ 0 & \text{otherwise.} \end{cases}$$

[1]:
```
import jax.numpy as jnp
def optimal_control(self, state, grad_value):
    y, v_y, phi, omega = state
    p1, p2, p3, p4 = grad_value

    coeff_T1 = p2 * jnp.cos(phi) / self.m - p4 * self.l / self.Iyy
    coeff_T2 = p2 * jnp.cos(phi) / self.m + p4 * self.l / self.Iyy

    T_1 = jnp.where(coeff_T1 < 0, self.max_thrust_per_prop, 0.0)
    T_2 = jnp.where(coeff_T2 < 0, self.max_thrust_per_prop, 0.0)

    return jnp.array([T_1, T_2])
```

(b)

The target set is $\mathcal{T} = [3, 7] \times [-1, 1] \times [-\pi/12, \pi/12] \times [-1, 1]$. We need $h(\mathbf{x}) \le 0 \iff \mathbf{x} \in \mathcal{T}$.

For each dimension, $x \in [a, b] \iff \max(a - x,\ x - b) \leq 0$. Taking the pointwise maximum over all dimensions:

$$h(\mathbf{x}) = \max\Big\{\max(3 - y,\ y - 7),\ \max(-1 - v_y,\ v_y - 1),\ \max(-\tfrac{\pi}{12} - \phi,\ \phi - \tfrac{\pi}{12}),\ \max(-1 - \omega,\ \omega - 1)\Big\}.$$

[2]:
```
def target_set(state):
    y, v_y, phi, omega = state
    return jnp.max(jnp.array([
        jnp.maximum(3.0 - y, y - 7.0),
        jnp.maximum(-1.0 - v_y, v_y - 1.0),
        jnp.maximum(-jnp.pi / 12.0 - phi, phi - jnp.pi / 12.0),
        jnp.maximum(-1.0 - omega, omega - 1.0),
    ]))
```

(c)

The operational envelope is $\mathcal{E} = [1, 9] \times [-6, 6] \times [-\infty, \infty] \times [-8, 8]$. Similarly to part (b):

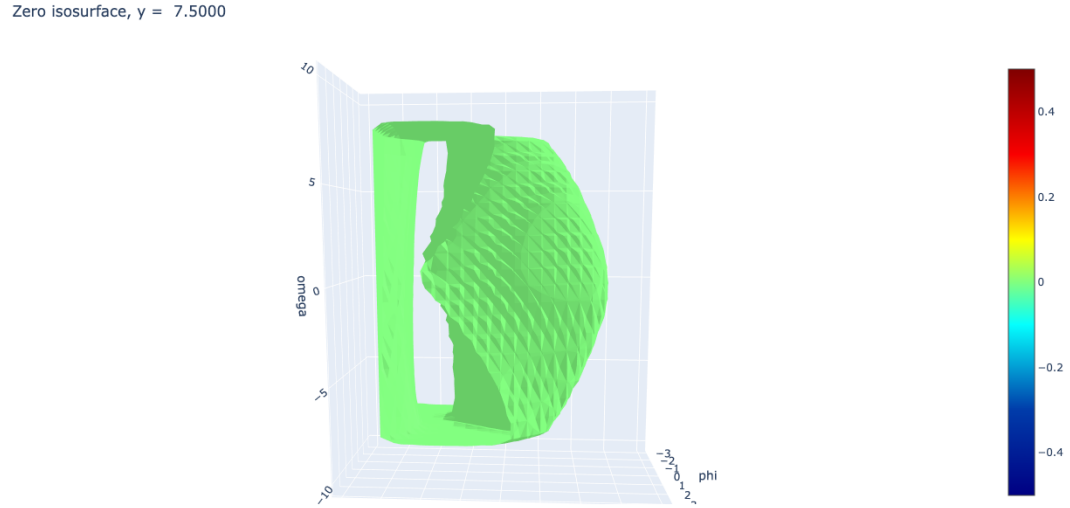
$$e(\mathbf{x}) = \max\Big\{\max(1 - y,\ y - 9),\ \max(-6 - v_y,\ v_y - 6),\ \max(-8 - \omega,\ \omega - 8)\Big\}.$$

e does not contain ϕ as there is no constraint on ϕ.

[3]:
```
def envelope_set(state):
    y, v_y, phi, omega = state
    return jnp.max(jnp.array([
        jnp.maximum(1.0 - y, y - 9.0),
        jnp.maximum(-6.0 - v_y, v_y - 6.0),
        jnp.maximum(-8.0 - omega, omega - 8.0),
    ]))
```

(d)

[4]:
```
from IPython.display import Image, display
display(Image(filename='isosurface.png', width=700))
```



At $y = 7.5$ we are near the top of the envelope ($y \leq 9$), so the enclosed region shows which (v_y, ϕ, ω) states can still reach the target without leaving $\mathcal{E}$.

The diagonal tilt in (ϕ, ω) comes from the fact that if ϕ and ω have opposite signs (say $\phi > 0, \omega < 0$), the pitch rate is already correcting the tilt, so recovery is easier. But if they have the same sign, the drone keeps rotating the wrong way and since thrust is bounded ($T_1, T_2 \in [0, T_{\max}]$) we can't reverse the rotation fast enough before leaving the envelope. Those corners get excluded from the recoverable set, which is why the surface stretches along the opposite-sign diagonal.

The surface is also not symmetric in v_y because at $y = 7.5$ an upward velocity only has 1.5m before hitting the ceiling at $y = 9$, while a downward velocity has 6.5m of room before the floor at $y = 1$. So the region is narrower for large positive v_y.

(e)

With DP/HJB we get a globally optimal policy over the whole state space, and at runtime it's just a table lookup so it's basically free to evaluate online. We also get the backward reachable set for free, which tells us exactly which states are recoverable. The main issue is the curse of dimensionality — the grid scales as $O(N^d)$ so both memory and offline compute blow up fast. That's why we had to drop from 6D to 4D here. Also, since the Hamiltonian is linear in the controls, the policy ends up being bang-bang ($T_1, T_2 \in \{0, T_{\max}\}$), which is optimal in theory but in practice means aggressive switching that's bad for actuators. And if we want to add obstacles later, we have to redo the whole computation.

MPC doesn't have the dimensionality problem since it just solves a local optimization at each step instead of gridding the state space. It's also easy to add new constraints or obstacles on the fly, and the optimization naturally gives smoother controls. The tradeoff is that it's only locally optimal, so it can fail from hard initial conditions where DP would have found a way out. Plus we need to solve an optimization at every timestep, which can be tight for something as fast as a self-righting maneuver.

Problem3

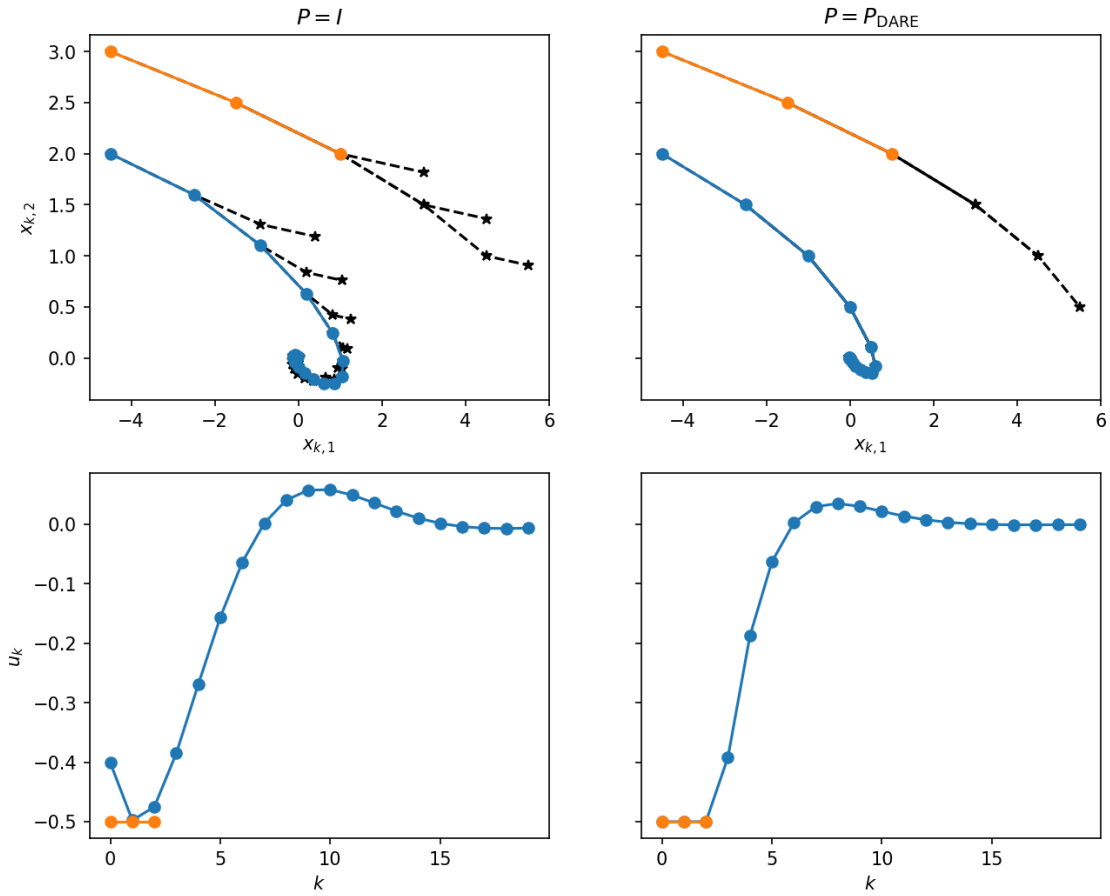
May 22, 2026

0.1 3.3 MPC feasibility

(a)

[1]:
```
import cvxpy as cvx
def do_mpc(x0, A, B, P, Q, R, N, rx, ru, rf):
    n, m = Q.shape[0], R.shape[0]
    x_cvx = cvx.Variable((N + 1, n))
    u_cvx = cvx.Variable((N, m))

    cost = cvx.quad_form(x_cvx[N], P)
    for k in range(N):
        cost += cvx.quad_form(x_cvx[k], Q) + cvx.quad_form(u_cvx[k], R)

    constraints = [x_cvx[0] == x0]
    for k in range(N):
        constraints += [
            x_cvx[k + 1] == A @ x_cvx[k] + B @ u_cvx[k],
            cvx.norm_inf(x_cvx[k]) <= rx,
            cvx.norm_inf(u_cvx[k]) <= ru,
        ]
    constraints += [cvx.norm_inf(x_cvx[N]) <= rf]

    prob = cvx.Problem(cvx.Minimize(cost), constraints)
    prob.solve(cvx.CLARABEL)
    return x_cvx.value, u_cvx.value, prob.status
```

[2]:
```
from IPython.display import Image, display
display(Image(filename='mpc_feasibility_sim.png', width=700))
```



For both values of P, the initial condition $x_0 = (-4.5, 2)$ (blue) converges to the origin. With $P = I$, convergence is slower and the predicted open-loop trajectories (dashed stars) diverge from the realized closed-loop trajectory. $P = I$ underestimates the true cost-to-go beyond the horizon, so the planner does not penalize the terminal state enough. With $P = P_{\text{DARE}}$, the predicted open-loop steps nearly coincide with the closed-loop trajectory (the stars overlap the dots), because P_{DARE} is the true infinite-horizon cost-to-go for the unconstrained LQR, so replanning produces essentially the same trajectory.

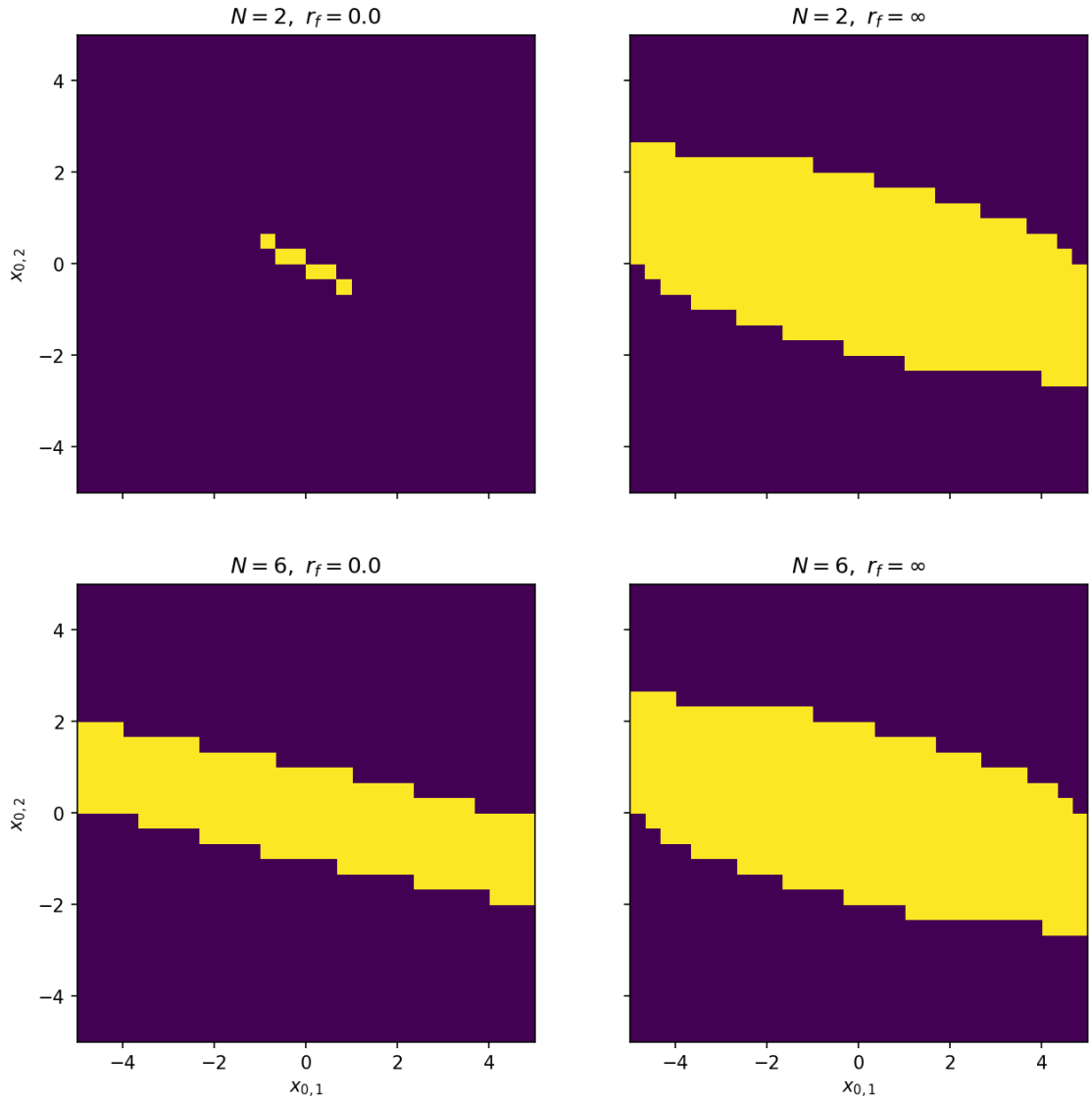
The initial condition $x_0 = (-4.5, 3)$ (orange) becomes infeasible at $t = 3$ for both values of P. The control saturates at $u = -0.5$ for all steps but the initial velocity $x_2 = 3$ is too large. The state drifts to $(3.0, 1.5)$ where MPC cannot find a feasible plan that satisfies $\|x_k\|_\infty \leq 5$ over the next $N = 3$ steps. This initial condition is outside the region of attraction for this MPC configuration regardless of the choice of P.

(b)

[3]:
```
import numpy as np
def compute_roa(A, B, P, Q, R, N, rx, ru, rf, grid_dim=21, max_steps=20,␣
 ↪tol=1e-2):
    roa = np.zeros((grid_dim, grid_dim))
```

```
    xs = np.linspace(-rx, rx, grid_dim)
    for i, x1 in enumerate(xs):
        for j, x2 in enumerate(xs):
            x = np.array([x1, x2])
            for _ in range(max_steps):
                x_mpc, u_mpc, status = do_mpc(x, A, B, P, Q, R, N, rx, ru, rf)
                if status == "infeasible":
                    break
                x = A @ x + B @ u_mpc[0, :]
                if np.linalg.norm(x) <= tol:
                    roa[i, j] = 1
                    break
    return roa
```

[4]:
```
from IPython.display import Image, display
display(Image(filename='mpc_feasibility_roa.png', width=700))
```



The ROA grows as we increase N and as we relax r_f.

With $r_f = 0$, the MPC must drive the state exactly to the origin within N steps. For $N = 2$ this is extremely restrictive since only a tiny set of initial states near the origin can reach $x = 0$ in 2 steps with $\|u\|_\infty \leq 0.5$, resulting in an almost empty ROA. Increasing to $N = 6$ gives the controller more steps to steer to the origin, so the ROA grows significantly.

With $r_f = \infty$, there is no terminal constraint so the MPC only needs to keep $\|x_k\|_\infty \leq r_x$ and $\|u_k\|_\infty \leq r_u$ over the horizon, which is a much weaker requirement and more states admit a feasible plan. However, without a terminal constraint there is no guarantee of persistent feasibility since after applying u_0 and re-solving from the new state, the next solve might become infeasible. With $r_f = 0$ this cannot happen because if the current plan reaches $x_N = 0$, we can always build a feasible plan from the next state by reusing the remaining controls and appending $u = 0$. So $r_f = 0$ gives a smaller but safe ROA where convergence is guaranteed, while $r_f = \infty$ gives a larger ROA

without that guarantee.

The diagonal shape of the ROA reflects the system dynamics since x_1 represents position and x_2 velocity. States where x_1 and x_2 have the same sign are harder to control back to the origin because the velocity pushes the position further away, while states where x_2 opposes x_1 are naturally returning.

Problem4

May 22, 2026

0.1 3.4 Terminal ingredients

(a)

$A = \begin{bmatrix} 0.9 & 0.6 \\ 0 & 0.8 \end{bmatrix}$ has eigenvalues 0.9 and 0.8, both strictly less than 1, so the uncontrolled system $x_{t+1} = Ax_t$ is asymptotically stable. This allows us to design $\mathcal{X}_T$ and P using only $x_{t+1} = Ax_t$ with $u = 0$.

Recursive feasibility requires $\mathcal{X}_T$ to be a positive invariant set for $x_{t+1} = Ax_t$ contained in $\mathcal{X} = \{x : \|x\|_2 \leq r_x\}$. We could take $\mathcal{X}_T = \mathcal{O}_\infty$, the maximal positive invariant set within $\mathcal{X}$.

To use the MPC stability theorem (for quadratic cost), we verify:

A0: $Q = Q^\top \succ 0$, $R = R^\top \succ 0$, $P \succ 0$. This holds since $Q = I$, $R = I$, and we choose $P \succ 0$.

A1: $\mathcal{X}$, $\mathcal{X}_T$, and $\mathcal{U}$ contain the origin in their interior and are closed. This holds by construction.

A2: $\mathcal{X}_T \subseteq \mathcal{X}$ is control invariant and bounded. Since $u = 0$ is a feasible control and $\mathcal{X}_T$ is positive invariant under $x_{t+1} = Ax_t$, it is control invariant for the full system.

A3: Since $u = 0$ is feasible and $Ax \in \mathcal{X}_T$ if $x \in \mathcal{X}_T$, assumption A3 becomes

$$-x^\top Px + x^\top Qx + x^\top A^\top PA\, x \leq 0, \quad \forall x \in \mathcal{X}_T,$$

which is true since, due to the fact that A is asymptotically stable,

$$\exists\, P \succ 0 \;\text{ s.t. }\; -P + Q + A^\top PA = 0 \quad \text{(discrete Lyapunov equation)}.$$

(b)

Let $x \in \mathcal{X}_T$, i.e., $x^\top Wx \leq 1$.

Positive invariance: since $A^\top WA - W \preceq 0$,

$$(Ax)^\top W(Ax) = x^\top A^\top WA\, x \leq x^\top Wx \leq 1,$$

so $Ax \in \mathcal{X}_T$. Hence $\mathcal{X}_T$ is positive invariant for $x_{t+1} = Ax_t$.

State constraints: since $I - r_x^2 W \preceq 0$,

$$x^\top (I - r_x^2 W)x \leq 0$$

$$\Rightarrow \|x\|_2^2 \leq r_x^2\, x^\top W x$$

Since $x \in \mathcal{X}_T$, $x^\top W x \leq 1$, so

$$\|x\|_2^2 \leq r_x^2 \cdot 1 = r_x^2$$
$$\Rightarrow \|x\|_2 \leq r_x$$
$$\Rightarrow \mathcal{X}_T \subseteq \mathcal{X} \quad \text{(respects state constraints).}$$

(c)

Let $M := W^{-1}$. Since W is symmetric positive definite, its eigenvalues $\lambda_i > 0$. The eigenvalues of $M = W^{-1}$ are $1/\lambda_i > 0$, and M is symmetric (the inverse of a symmetric matrix is symmetric), so M is also symmetric positive definite. Reciprocally, $M \succ 0 \Rightarrow W = M^{-1} \succ 0$ by the same argument.

Constraint 1: $A^\top W A - W \preceq 0$ with $W = M^{-1}$:

$$A^\top M^{-1} A \preceq M^{-1} \iff M^{-1} - A^\top M^{-1} A \succeq 0$$

Using hint 1 (pre/post multiply by $M \succ 0$):

$$M(M^{-1} - A^\top M^{-1} A)M = M - (AM)^\top M^{-1}(AM) \succeq 0$$

By Schur's complement lemma:

$$\begin{bmatrix} M & MA^\top \\ AM & M \end{bmatrix} \succeq 0$$

Constraint 2: $I - r_x^2 W \preceq 0$ with $W = M^{-1}$:

$$r_x^2 M^{-1} \succeq I \iff M^{-1} \succeq \frac{1}{r_x^2} I$$

By hint 2 ($A^{-1} \succeq \frac{1}{\gamma} I \iff A \preceq \gamma I$):

$$M \preceq r_x^2 I \iff r_x^2 I - M \succeq 0$$

Objective: $\log\det(W^{-1}) = \log\det(M)$, which is concave in $M \succ 0$.

The reformulated concave SDP is:

$$\begin{aligned} \underset{M \succ 0}{\text{maximize}} \quad & \log\det(M) \\ \text{subject to} \quad & \begin{bmatrix} M & MA^\top \\ AM & M \end{bmatrix} \succeq 0 \end{aligned}$$

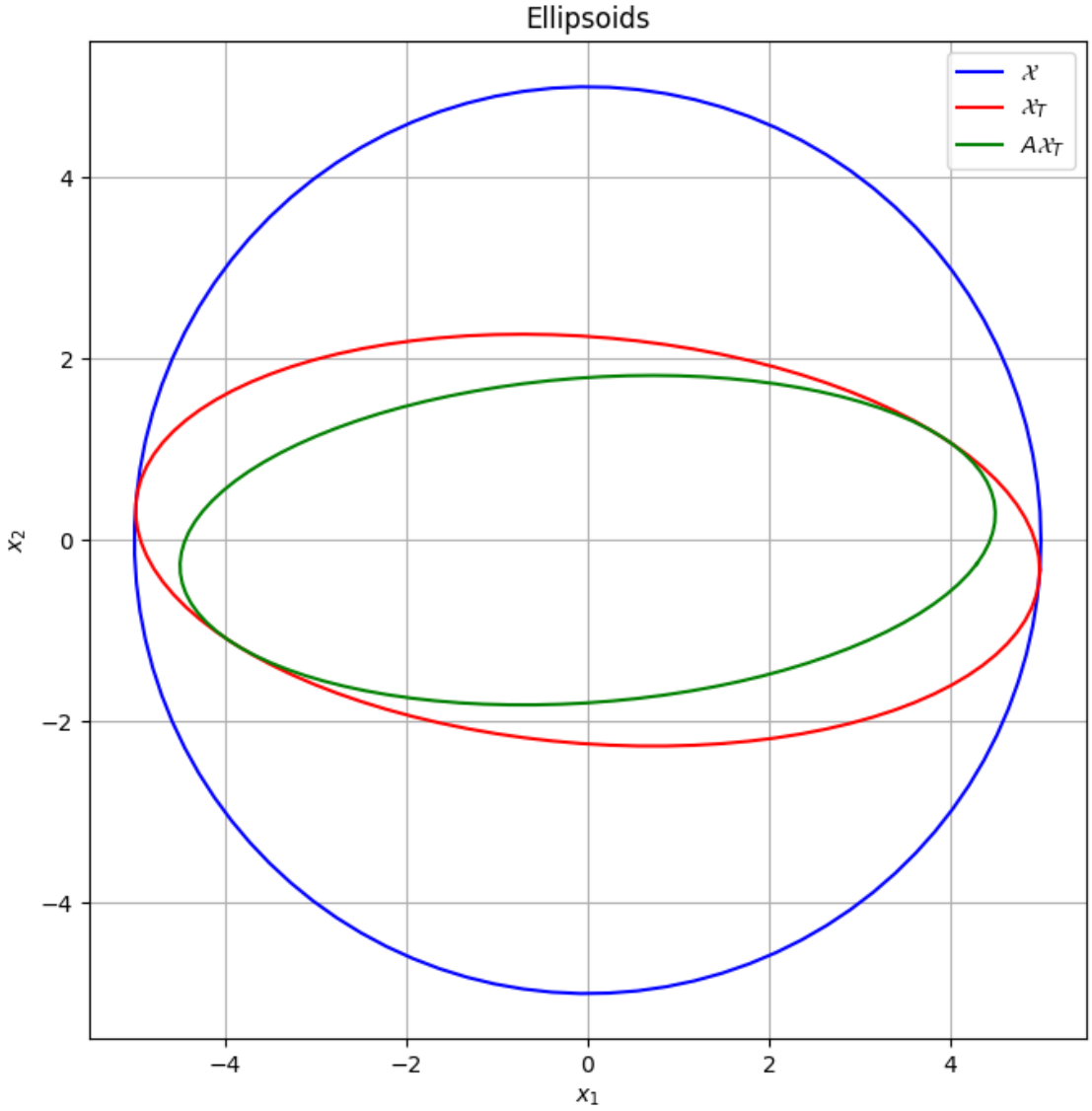
$$r_x^2 I - M \succeq 0$$

(d)

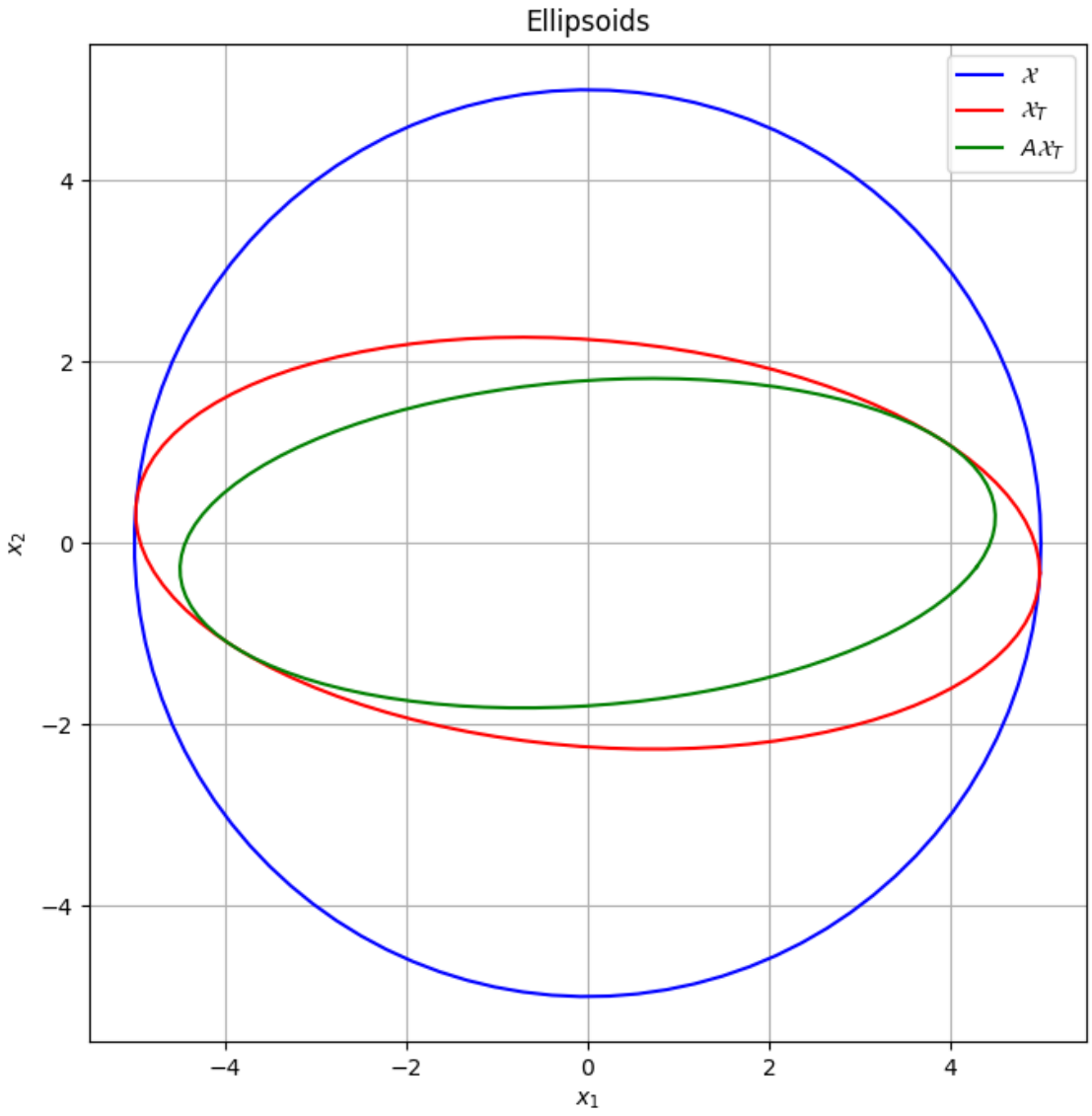
[1]:
```
import numpy as np
import cvxpy as cvx
import matplotlib.pyplot as plt

A = np.array([[0.9, 0.6], [0.0, 0.8]])
B = np.array([[0.0], [1.0]])
n = 2
rx = 5.0
ru = 1.0
Q = np.eye(n)
R = np.eye(1)
P = np.eye(n)
N = 4

# Solve the SDP for M
M = cvx.Variable((n, n), symmetric=True)
constraints = [
    cvx.bmat([[M, M @ A.T], [A @ M, M]]) >> 0,
    rx**2 * np.eye(n) - M >> 0,
]
prob = cvx.Problem(cvx.Maximize(cvx.log_det(M)), constraints)
prob.solve(cvx.SCS)

M_val = M.value
W_val = np.linalg.inv(M_val)
print(f"W = M^{{-1}} =\n{np.round(W_val, 3)}")

def generate_ellipsoid_points(M, num_points=100):
    L = np.linalg.cholesky(M)
    theta = np.linspace(0, 2 * np.pi, num_points)
    u = np.column_stack([np.cos(theta), np.sin(theta)])
    x = u @ L.T
    return x

X_pts = generate_ellipsoid_points(rx**2 * np.eye(n))
XT_pts = generate_ellipsoid_points(M_val)
AXT_pts = (A @ XT_pts.T).T

fig, ax = plt.subplots(figsize=(8, 8))
ax.plot(X_pts[:, 0], X_pts[:, 1], 'b-', label=r'$\mathcal{X}$')
ax.plot(XT_pts[:, 0], XT_pts[:, 1], 'r-', label=r'$\mathcal{X}_T$')
ax.plot(AXT_pts[:, 0], AXT_pts[:, 1], 'g-', label=r'$A\mathcal{X}_T$')
ax.set_xlabel(r'$x_1$')
ax.set_ylabel(r'$x_2$')
```

```
ax.set_title('Ellipsoids')
ax.set_aspect('equal')
ax.legend()
ax.grid(True)
plt.show()
```

```
W = M^{-1} =
[[0.041 0.013]
 [0.013 0.198]]
```



[2]:
```
from IPython.display import Image, display
display(Image(filename='ellipsoids.png', width=600))
```

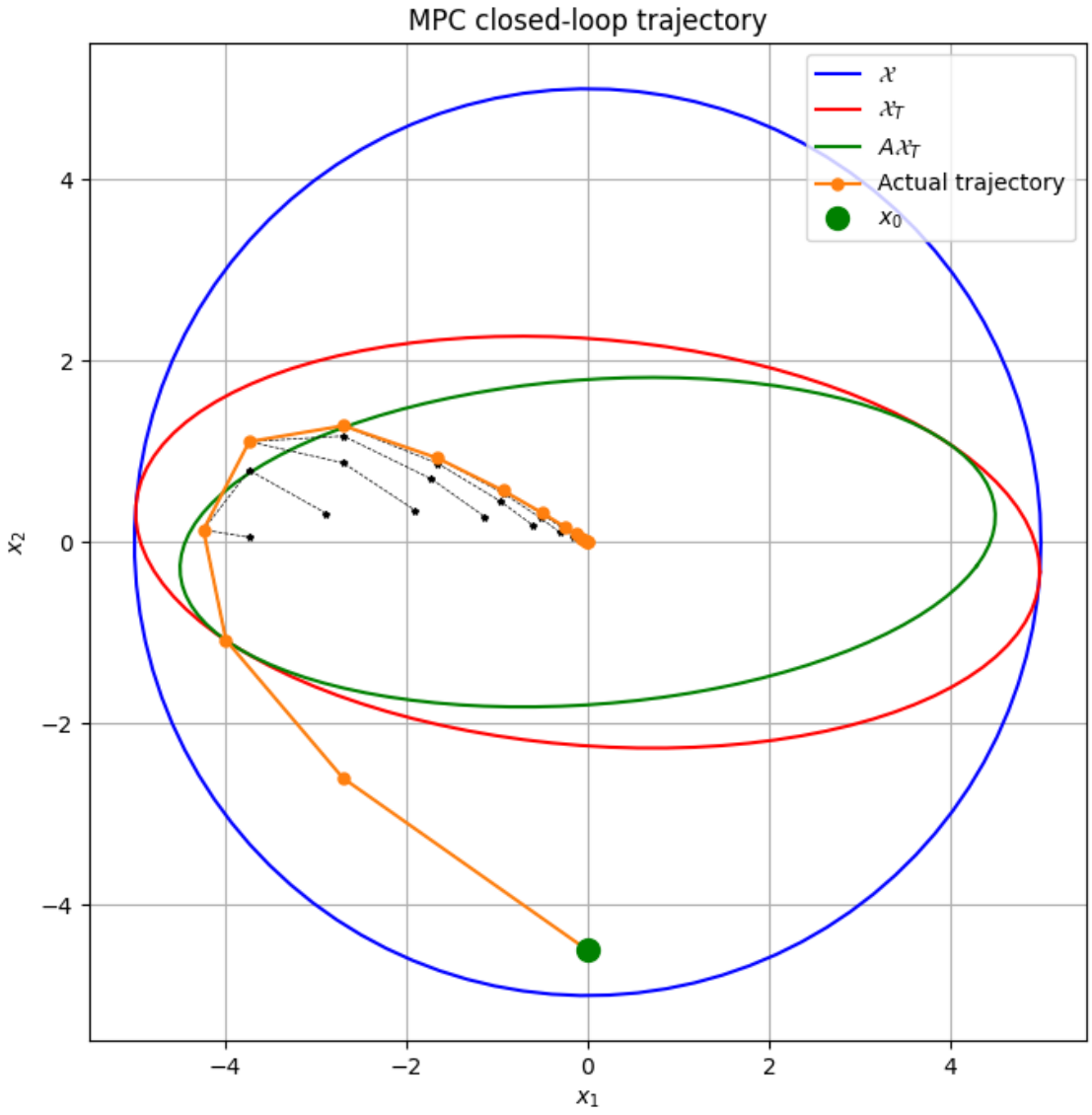


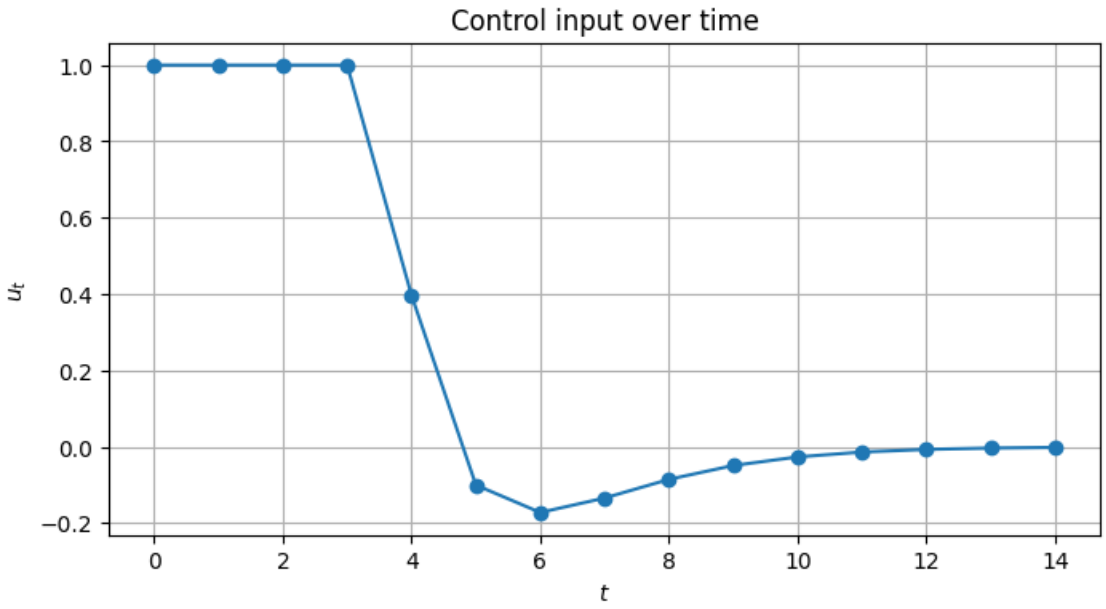
(e)

[3]:
```
# Parameterized MPC problem
x0_param = cvx.Parameter(n)
x_mpc = cvx.Variable((N + 1, n))
u_mpc = cvx.Variable((N, 1))

cost = cvx.quad_form(x_mpc[N], P)
for k in range(N):
    cost += cvx.quad_form(x_mpc[k], Q) + cvx.quad_form(u_mpc[k], R)

mpc_constraints = [x_mpc[0] == x0_param]
```

```
for k in range(N):
    mpc_constraints += [
        x_mpc[k + 1] == A @ x_mpc[k] + B @ u_mpc[k],
        cvx.norm2(x_mpc[k]) <= rx,
        cvx.norm2(u_mpc[k]) <= ru,
    ]
mpc_constraints += [cvx.quad_form(x_mpc[N], W_val) <= 1]

mpc_prob = cvx.Problem(cvx.Minimize(cost), mpc_constraints)

# Simulate from x0 = (0, -4.5) for 15 steps
T_sim = 15
x0 = np.array([0.0, -4.5])
x_actual = np.zeros((T_sim + 1, n))
u_actual = np.zeros((T_sim, 1))
x_planned = []

x_actual[0] = x0
for t in range(T_sim):
    x0_param.value = x_actual[t]
    mpc_prob.solve(cvx.CLARABEL)
    x_planned.append(x_mpc.value.copy())
    u_actual[t] = u_mpc.value[0, :]
    x_actual[t + 1] = A @ x_actual[t] + B.flatten() * u_actual[t, 0]

# State trajectories overlaid on ellipsoids
fig, ax = plt.subplots(figsize=(8, 8))
ax.plot(X_pts[:, 0], X_pts[:, 1], 'b-', label=r'$\mathcal{X}$')
ax.plot(XT_pts[:, 0], XT_pts[:, 1], 'r-', label=r'$\mathcal{X}_T$')
ax.plot(AXT_pts[:, 0], AXT_pts[:, 1], 'g-', label=r'$A\mathcal{X}_T$')
for t in range(T_sim):
    ax.plot(x_planned[t][:, 0], x_planned[t][:, 1], '--*', color='k',␣
 ↪markersize=3, linewidth=0.5)
ax.plot(x_actual[:, 0], x_actual[:, 1], '-o', color='tab:orange', markersize=5,␣
 ↪label='Actual trajectory')
ax.plot(x_actual[0, 0], x_actual[0, 1], 'go', markersize=10, label=r'$x_0$')
ax.set_xlabel(r'$x_1$')
ax.set_ylabel(r'$x_2$')
ax.set_title('MPC closed-loop trajectory')
ax.set_aspect('equal')
ax.legend()
ax.grid(True)
plt.show()

# Control input over time
fig, ax = plt.subplots(figsize=(8, 4))
ax.plot(range(T_sim), u_actual[:, 0], '-o')
```

```
ax.set_xlabel(r'$t$')
ax.set_ylabel(r'$u_t$')
ax.set_title('Control input over time')
ax.grid(True)
plt.show()
```




Control input over time
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
u_t
0
2
4
6
8
10
12
14
t